

Supplementary information

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This supplement accompanies the manuscript “Matched nulls and compact baselines qualify biochemical representations for metabolite reconstruction”. All model outcomes, including negative controls and implementation failures, are retained. Primary results use only the accepted human revision-2 and full CCLE runs.

S1. Cohorts, masks and independent units

ST002081 uses 1,539 assay records from 112 repository participant identifiers after removing 104 QC/missing-identifier rows. The final Hornburg et al. 2023 publication independently reports 1,539 samples, 112 participants and 104 QC samples. The deposit narrative says 1,546/109; its history is not explained by the available records. Repeated visits remain together. ST000818 provides 450 unique sample identifiers and 15 population categories and is described by its source as 450 individuals; distinct donor identity was not independently verified. CCLE aligns 913 lines but evaluates 876 lines in 18 retained lineages for every one of 60 targets. Each target has two predictions per line across two outer repeats, giving 105,120 prediction rows per model; these are not 105,120 independent cell lines.

The human primary panels contain 493 and 255 complete lipid labels. Masks are family-stratified and disjoint. The human structural representation has 95 and 81 descriptors before visible-mask restriction. Representation rules are refit independently within each cohort; no frozen ST002081 model is transferred to ST000818. CCLE uses 76 mapped metabolites and 317 direct-reaction candidate rows regenerated from Human-GEM v2.0.0. Each target is excluded from its own inputs. Its compact neighbor panel contains 1–19 metabolites, median 3; correlation selection uses the same target-specific prediction-time count while inspecting 75 candidate inputs during training.

Ordinary rows are equally weighted for fitting. Reported human evaluation first averages hidden-target errors within each sample/mask, then masks within samples, samples within participants, and finally participants equally; ST000818 uses population categories at the final level. CCLE averages repeated/cell-line losses within each target/lineage, then available lineages and fixed targets equally before taking a square root. Biological-group bootstrap intervals condition on fitted models and the fixed target panel. The inferential claim does not include refitting variability or unidentified future populations.

S2. Complete primary scorecards

ST002081

Model	Primary RMSE	Row-weighted RMSE
all visible ridge	0.1780	0.1747
degree preserving null	0.4192	0.4096
descriptor local svd	0.2434	0.2389
family median	0.6680	0.6503

Model	Primary RMSE	Row-weighted RMSE
pca matched dimension	0.1964	0.1929
population mean	1.0652	1.0129
structural median	0.3168	0.3091

ST000818

Model	Primary RMSE	Row-weighted RMSE
all visible ridge	0.8491	0.8457
degree preserving null	1.8030	1.7873
descriptor local svd	0.9574	0.9543
family median	1.8869	1.8714
pca matched dimension	0.8394	0.8376
population mean	2.0130	1.9983
structural median	1.2267	1.2201

CCLE

Model	Primary RMSE	Row-weighted RMSE
all other metabolites ridge	0.6648	0.7014
correlation selected ridge	0.7315	0.7684
direct neighbor ridge	0.8239	0.8696
global pca ridge	0.8090	0.8412
network additive ridge	0.8117	0.8639
network interaction ridge	0.8170	0.8748
population mean	0.9865	1.0666

S3. Prespecified primary contrasts

Positive improvement is statistical-reference RMSE minus mechanism-representation RMSE. Negative results favor the reference. All five contrasts were defined before extension outcomes. Descriptive 95% intervals and 99% per-contrast intervals are retained in the source CSV. The latter target nominal 95% Bonferroni familywise coverage across five contrasts, subject to the conditional bootstrap assumptions. The table reports 99% per-contrast limits. Main Figure 3 uses thick 95% and thin 99% per-contrast intervals. All five two-sided Monte Carlo sign-flip tests have raw $p = 0.0000999900$ and Holm-adjusted $p = 0.0004999500$ across the five tests. The raw probabilities reach the 1/10,001 simulation-resolution floor, not an arbitrarily small exact p value. Sign flips assume exchangeability of paired group-effect signs under the null. Overlapping cross-validation training sets can induce dependence across held-out groups that these procedures do not fully represent. Inference remains conditional on the fitted models and fixed target panel, without full-pipeline refitting.

Dataset	Reference	Improvement	99% lower	99% upper
st002081	pca matched dimension	-0.1204	-0.1278	-0.1140

Dataset	Reference	Improvement	99% lower	99% upper
st002081	descriptor local svd	-0.0733	-0.0789	-0.0676
st000818	pca matched dimension	-0.3873	-0.7902	-0.0712
st000818	descriptor local svd	-0.2693	-0.5388	-0.0494
ccle	correlation selected ridge	-0.0924	-0.1073	-0.0807

S4. Information and fitting budgets

Measured inputs, output dimension, learned transform loadings, ridge coefficients and tuning budgets are separate. Human models generally see the same visible assay panel; 95 descriptors do not mean a 95-analyte panel. PCA has learned loadings not required by the fixed median membership. Below, transform loadings exclude common centering/scaling parameters and intercepts; total parameter equality is not claimed. Every fitted ridge model uses the same four-alpha grid and three grouped inner folds. Both PC representations are training-standardized before ridge.

ST002081 budgets

Model	Visible inputs	Dimension	Median loadings
all visible ridge	390-397	390-397	0
degree preserving null	390-397	95-95	0
descriptor local svd	390-397	95-95	2798
family median	390-397	9-9	0
pca matched dimension	390-397	95-95	37525
structural median	390-397	95-95	0

ST000818 budgets

Model	Visible inputs	Dimension	Median loadings
all visible ridge	202-208	202-208	0
degree preserving null	202-208	81-81	0
descriptor local svd	202-208	81-81	917
family median	202-208	6-6	0
pca matched dimension	202-208	81-81	16443
structural median	202-208	81-81	0

CCLE budgets

Model	Median metabolites	Median genes	Median dimension	Selection pool
all other metabolites ridge	75.0	0.0	75.0	0.0
correlation selected ridge	3.0	0.0	3.0	75.0
direct neighbor ridge	3.0	0.0	3.0	0.0
global pca ridge	75.0	0.0	3.0	0.0
network additive ridge	3.0	5.0	8.0	0.0
network interaction ridge	3.0	5.0	13.0	0.0

Model	Median metabolites	Median genes	Median dimension	Selection pool
population mean	0.0	0.0	0.0	0.0

Correlation-selected panels use 75 candidate metabolites in training and 1-19 at prediction. This comparison matches deployment metabolite count, not development assay cost. The direct-neighbor map instead uses curated network information. PCA sees all 75 non-target metabolites even when only a few components are retained. No assay-cost, user-interpretability or clinical-benefit experiment was performed.

S5. Full human sensitivity scorecards

These sensitivities are descriptive, not additional primary discoveries. Training-only missingness starts from all eligible assay labels, retains visible inputs with $\geq 80\%$ training observations, and keeps only targets complete and variable in the applicable training partition. Inner target/feature rules use inner training only. Test errors include observed targets only. Changing the eligible universe changes the task; differences across modes are not effects of a model change on identical targets. In ST000818 the missingness version retains the same 255-feature panel and is numerically identical to primary, so it provides no additional missingness stress test. Resolution sensitivity removes unsupported individual-chain memberships from sum-composition-only labels while retaining reported family and total composition. It is a conservative name-parser check, not expert reannotation.

ST002081 — missingness

Schema: 841 features; 135 descriptors. Retained targets per outer mask/fold: 96-109. Full fold exclusions, changing visible panels, observed-target denominators and budgets are retained in run CSVs.

Model	Primary RMSE
all visible ridge	0.1860
degree preserving null	0.3995
descriptor local svd	0.2409
family median	0.6599
pca matched dimension	0.2137
population mean	1.0650
structural median	0.3100

ST002081 — resolution

Schema: 493 features; 95 descriptors. Retained targets per outer mask/fold: 96-103. Full fold exclusions, changing visible panels, observed-target denominators and budgets are retained in run CSVs.

Model	Primary RMSE
all visible ridge	0.1780
degree preserving null	0.4153
descriptor local svd	0.2433
family median	0.6680

Model	Primary RMSE
pca matched dimension	0.1964
population mean	1.0652
structural median	0.3163

ST000818 — missingness

Schema: 255 features; 81 descriptors. Retained targets per outer mask/fold: 47–53. Full fold exclusions, changing visible panels, observed-target denominators and budgets are retained in run CSVs.

Model	Primary RMSE
all visible ridge	0.8491
degree preserving null	1.8030
descriptor local svd	0.9574
family median	1.8869
pca matched dimension	0.8394
population mean	2.0130
structural median	1.2267

ST000818 — resolution

Schema: 255 features; 67 descriptors. Retained targets per outer mask/fold: 47–53. Full fold exclusions, changing visible panels, observed-target denominators and budgets are retained in run CSVs.

Model	Primary RMSE
all visible ridge	0.8491
degree preserving null	1.7860
descriptor local svd	0.9619
family median	1.8869
pca matched dimension	0.8630
population mean	2.0130
structural median	1.2347

S6. Outcome-aware secondary mean aggregator

An AI-assisted supervisory review requested the inexpensive mean aggregator proposed in the initial literature audit. This amendment was recorded after the five primary outcomes were viewed; it adds no primary test. Arithmetic means use exactly the same visible descriptor memberships and train-only standardization, nested tuning and outer splits as the median/SVD comparison. All seven primary models are rerun alongside it. One startup failed before predictions because a read-only NumPy view was modified in place; the failure and repair are retained.

ST002081 — all secondary-run models

Model	Primary RMSE
all visible ridge	0.1780

Model	Primary RMSE
degree preserving null	0.4192
descriptor local svd	0.2434
descriptor mean	0.2425
family median	0.6680
pca matched dimension	0.1964
population mean	1.0652
structural median	0.3168

ST000818 — all secondary-run models

Model	Primary RMSE
all visible ridge	0.8491
degree preserving null	1.8030
descriptor local svd	0.9574
descriptor mean	0.9443
family median	1.8869
pca matched dimension	0.8394
population mean	2.0130
structural median	1.2267

S7. Reproduction, retained nulls and sensitivity grids

Fourteen released pipeline/report stages completed independently in the new environment. 51 of 62 declared outputs are byte-identical. All 14 manifest decisions agree. The remaining 11 mismatches include ST002081 all-visible ridge numerics and their downstream tables, report and figure consequences. Largest coarse summary RMSE difference is 0.000002997 SD, largest target RMSE difference approximately 0.000475 SD; two sample-level priority rankings change precision by 1/3. No tolerance was used to call these files byte-identical. Exact changed-value counts, maximum absolute/relative differences, interval changes and gate comparisons are in the audit CSVs. The cause is not established by package-version identity. The released source bytes match the public ZIP, but three historical implementation digests cannot be recovered from available copies.

Stage	Outputs	Byte-identical	Decision agrees
pathway-score-st002081	7	2	True
pathway-score-st002081-structural	6	3	True
pathway-score-st000818-replication	6	6	True
pathway-score-ccle	5	5	True
pathway-score-st002081-structural-null-sensitivity	2	2	True
pathway-score-ccle-null-ensemble	3	3	True
pathway-score-ccle-property-matched-null	3	3	True
pathway-score-human-sensitivity	3	3	True

Stage	Outputs	Byte-identical	Decision agrees
pathway-score-human-graph-mixing	3	3	True
pathway-score-ccle-sensitivity	3	3	True
pathway-score-ccle-reaction-cluster	4	4	True
pathway-score-mapping-supplement	6	6	True
pathway-score-publication-figures	7	5	True
factorized-pathway-publication	4	3	True

Complete archived and regenerated null grids are included in the public code archive under evidence/baseline and evidence/new: 20 human structural nulls, 20 CCLE dimension-matched draws, 20 CCLE degree/coverage-matched draws, and ST000818's 20 nulls. Nine human sensitivity settings per cohort, ten CCLE settings, graph-mixing diagnostics, mapping ledgers and reaction-subsystem robustness tables are retained. These grids were originally adaptive; passing all sampled null realizations is not an exact graph-randomization p value or proof of uniform graph sampling. Archived CCLE target-bootstrap intervals remain a distinct estimand from the new biological-lineage intervals.

S8. Audit and version history

AUD-01-14 dispositions are recorded in readiness/audit-resolution.csv. Repairs in the new analysis include common ST000818 splits, one baseline per observation, group-level uncertainty, corrected mean-imputation Methods, train-only missingness rules, explicit skill-statistic terminology, saved losses/splits and input preflight. The initial human extension accidentally weighted inner folds rather than groups, capped PCA by a rank upper bound, inherited outer target eligibility in inner validation, and aggregated missing masks before samples. Those outputs are preserved but superseded. Revision 2 fixes all four and passes nine targeted tests; CCLE passes ten tests. Independent arithmetic review agrees within 4.44e-16 and confirms all prediction and split identities. The CCLE numerical reviewer also wrote its implementation; that role is disclosed. A separate AI-assisted verification pass is likewise disclosed; it is not independent human peer review.

S9. Interpretation and rights

Chemical or reaction membership remains inspectable, but interpretability benefit was not measured. Success against matched random controls does not establish greater prediction accuracy, pathway causality, physiological flux, treatment response, rejuvenation or clinical value. Different cohorts are never pooled as if jointly measured; the human lipid cohorts do not provide same-person multimodal validation in this work. Genetic/topology, constraint and perturbational artifacts from the older package remain supporting/exploratory material and are not new independent confirmation here.

Both Workbench study APIs report CC BY 4.0; Human-GEM v2.0.0 is CC BY 4.0. Raw CCLE files are not redistributed and should be obtained from the original release [12]. The existing software archive is Apache-2.0 with separate content licensing. MIRTH was not executed or adapted because posted permissions are unresolved. Participant-

linked losses and human split maps are not redistributed. Authorship, funding, competing-interest and ethics statements appear in the main manuscript.

S10. Machine-readable evidence index

The public code archive includes results/ CSVs, figure source data, exact run manifests, code, configuration snapshots, frozen environment versions, defect dispositions and reproducibility instructions. Participant-level records and raw matrices are excluded. Detailed tuning/exclusion/budget grids are supplied electronically rather than rendered as tens of thousands of PDF rows. Captions and main tables link to the same result sources; no hand-entered alternative scorecard is used.

S11. Qualification ladder source values

Main Figure 1 uses current estimators throughout. Rungs are primary equal-group or equal-lineage RMSEs; matched nulls are expected values over 20 realizations (Section S12). Compact steps are the locked primary contrasts. Shares are descriptive point-estimate partitions of the training-mean-to-compact gap on the RMSE and MSE scales and have no intervals. The archived CCLE ensemble row is listed for completeness only; it used the released estimator and target-level resampling.

Dataset	Rung	Construction	Primary RMSE
ST002081	Training mean	no structure	1.0652
ST002081	Matched null	same shape, rewired membership	0.4168
ST002081	Biochemical descriptors	declared membership, median	0.3168
ST002081	Compact statistics	PCA, same dimension	0.1964
ST002081	Full-panel ridge	all visible inputs	0.1780
ST000818	Training mean	no structure	2.0130
ST000818	Matched null	same shape, rewired membership	1.7901
ST000818	Biochemical descriptors	declared membership, median	1.2267
ST000818	Compact statistics	PCA, same dimension	0.8394
ST000818	Full-panel ridge	all visible inputs	0.8491
CCLE	Training mean	no structure	0.9865
CCLE	Matched null	property-matched random neighbours	0.9238
CCLE	Reaction neighbours	declared neighbours	0.8239
CCLE	Compact statistics	correlation-selected, same count	0.7315
CCLE	Full-panel ridge	all other metabolites	0.6648

Dataset	Step	Estimate	95% lower	95% upper	Estimator
ST002081	biochemistry vs matched null	0.1000	0.0949	0.1053	expected over 20 null realizations; current estimator; biological-group bootstrap
ST002081	compact vs biochemistry	0.1204	0.1153	0.1258	locked primary contrast; biological-group bootstrap
ST002081	matched null share rmse	0.7464	—	—	descriptive share of training-mean- to-compact gap (RMSE scale)
ST002081	biochemistry share rmse	0.1151	—	—	descriptive share of training-mean- to-compact gap (RMSE scale)
ST002081	compact share rmse	0.1385	—	—	descriptive share of training-mean- to-compact gap (RMSE scale)
ST002081	matched null share mse	0.8767	—	—	descriptive share of training-mean- to-compact gap (MSE scale)
ST002081	biochemistry share mse	0.0669	—	—	descriptive share of training-mean- to-compact gap (MSE scale)
ST002081	compact share mse	0.0564	—	—	descriptive share of training-mean- to-compact gap (MSE scale)
ST000818	biochemistry vs matched null	0.5634	0.0463	1.0098	expected over 20 null realizations; current estimator; biological-group bootstrap
ST000818	compact vs biochemistry	0.3873	0.0748	0.6872	locked primary contrast; biological-group bootstrap
ST000818	matched null share rmse	0.1900	—	—	descriptive share of training-mean- to-compact gap (RMSE scale)

Dataset	Step	Estimate	95% lower	95% upper	Estimator
ST000818	biochemistry share rmse	0.4800	—	—	descriptive share of training-mean- to-compact gap (RMSE scale)
ST000818	compact share rmse	0.3300	—	—	descriptive share of training-mean- to-compact gap (RMSE scale)
ST000818	matched null share mse	0.2533	—	—	descriptive share of training-mean- to-compact gap (MSE scale)
ST000818	biochemistry share mse	0.5077	—	—	descriptive share of training-mean- to-compact gap (MSE scale)
ST000818	compact share mse	0.2390	—	—	descriptive share of training-mean- to-compact gap (MSE scale)
CCLE	biochemistry vs matched null	0.1000	0.0846	0.1197	expected over 20 null realizations; current estimator; biological-group bootstrap
CCLE	compact vs biochemistry	0.0924	0.0830	0.1034	locked primary contrast; biological-group bootstrap
CCLE	matched null share rmse	0.2457	—	—	descriptive share of training-mean- to-compact gap (RMSE scale)
CCLE	biochemistry share rmse	0.3920	—	—	descriptive share of training-mean- to-compact gap (RMSE scale)
CCLE	compact share rmse	0.3623	—	—	descriptive share of training-mean- to-compact gap (RMSE scale)
CCLE	matched null share mse	0.2732	—	—	descriptive share of training-mean- to-compact gap (MSE scale)
CCLE	biochemistry share mse	0.3988	—	—	descriptive share of training-mean- to-compact gap (MSE scale)

Dataset	Step	Estimate	95% lower	95% upper	Estimator
CCLE	compact share mse	0.3280	—	—	descriptive share of training-mean-to-compact gap (MSE scale)
CCLE	archived biochemistry vs matched null	0.1374	0.0971	0.1807	archived property-matched ensemble; released estimator; target bootstrap (supplement only)

S12. Post-review analyses (outcome-aware)

Both analyses were frozen in readiness/post-review-r1-amendment.md before their outcomes were computed and reuse the accepted code, splits and tuning. They do not alter the five primary contrasts.

Membership by aggregation (human cohorts)

Membership effect is expected-null RMSE minus declared RMSE over 20 degree-preserving null realizations; realization 0 reproduces the primary null exactly.

Dataset	Rule	Declared	Expected null	Effect	95% interval
ST002081	median	0.3168	0.4168	0.1000	0.095 to 0.105
ST002081	mean	0.2425	0.2417	−0.0009	−0.003 to 0.001
ST002081	local svd	0.2434	0.2789	0.0355	0.033 to 0.038
ST000818	median	1.2267	1.7901	0.5634	0.046 to 1.010
ST000818	mean	0.9443	1.0592	0.1149	0.012 to 0.216
ST000818	local svd	0.9574	1.3620	0.4046	0.019 to 0.764

Dataset	Rule	Positive effects, n/20	95% intervals above zero, n/20	Smallest effect	Largest effect
ST002081	median	20/20	20/20	0.0956	0.1050
ST002081	mean	5/20	0/20	−0.0025	0.0009
ST002081	local svd	20/20	20/20	0.0304	0.0409
ST000818	median	20/20	20/20	0.4488	0.5840
ST000818	mean	20/20	20/20	0.0341	0.1717
ST000818	local svd	20/20	20/20	0.1279	0.5898

CCLE matched null under the current estimator

Declared direct neighbours: 0.8239; expected property-matched null over 20 seeds: 0.9238; effect 0.1000 (lineage-bootstrap 95% interval 0.0846 to 0.1197). All 20 seeds had intervals above zero. The table below reports seed-specific results under the

current estimator. The archived ensemble, reported separately in Section S11, used the released estimator and target-level resampling and is not combined with the current ladder.

Seed	Effect	95% lower	95% upper
20262100	0.0980	0.0824	0.1177
20262101	0.0983	0.0825	0.1195
20262102	0.0918	0.0778	0.1091
20262103	0.1055	0.0898	0.1249
20262104	0.1036	0.0878	0.1238
20262105	0.1048	0.0883	0.1261
20262106	0.0906	0.0758	0.1094
20262107	0.1057	0.0893	0.1257
20262108	0.0980	0.0827	0.1173
20262109	0.0893	0.0748	0.1082
20262110	0.1070	0.0901	0.1305
20262111	0.1123	0.0957	0.1338
20262112	0.1019	0.0854	0.1225
20262113	0.0803	0.0670	0.0959
20262114	0.1066	0.0890	0.1299
20262115	0.0946	0.0785	0.1150
20262116	0.1008	0.0872	0.1172
20262117	0.1076	0.0896	0.1308
20262118	0.1030	0.0875	0.1234
20262119	0.0994	0.0850	0.1181

S13. Availability inventory

Material	Status	Location
Code, tests, figure scripts, frozen amendments	Public	code archive: scripts/, tests/, readiness/
Aggregate model metrics, contrasts, budgets, ladder and post-review summaries	Public	results/
Per-run summaries, tuning and budget grids, null grids (archived and regenerated)	Public	evidence/new, evidence/baseline, evidence/archived-ccle-property-matched-null
Raw Metabolomics Workbench and CCLE matrices	Not redistributed; regenerable	original repositories (Data availability)
Individual predictions, participant-linked losses and split maps	Withheld	regenerable locally from code and public source data